

Research Papers Reading Guide

Blockchain Genome Synthesis — Literature Review

Sage Clokey — BME 129C Capstone — Spring 2026

1. L-SCRaMbLE — Large-Scale Genome Rearrangements (2022)

2022.12.12.519280v1.full.pdf

Light-inducible Cre recombinase generates large-scale deletions, inversions, and translocations on synthetic yeast chromosomes, tunable by light exposure duration. Introduces “loxTag” genotyping to detect structural variants and uses Nanopore long-read sequencing to map complex karyotype changes. Circular synthetic chromosomes showed more diverse rearrangements than linear ones. **Focus on:** How SCRaMbLE provides a controllable, post-assembly diversification mechanism — the genome rearranges itself and selection finds what works.

2. Iterative SCRaMbLE (2024)

2024.12.06.627136v1.full.pdf

Applies repeated rounds of SCRaMbLE under selection to rescue poorly performing synthetic gene modules (refactored histidine biosynthesis). Developed SCOUT, a fluorescence reporter for sorting rearranged cells, and POLAR-seq for mapping genotype-phenotype relationships at high throughput. Key finding: improvements plateau at local maxima — rearrangement alone is not unlimited. **Focus on:** How iterative rearrangement + selection is essentially “mining” — the cell discovers functional gene arrangements through trial, but needs guided starting points.

3. Cell-Free Partial Genome Assembly (2025)

2025.10.16.682769v1.full.pdf

Assembled a 41-kb partial minimal genome (15 genes for phospholipid biosynthesis, DNA replication, cell division) from 14 DNA fragments using yeast homologous recombination, then expressed all proteins in the PURE cell-free system and replicated the genome in liposomes using phi29 polymerase. **Focus on:** Modular one-pot assembly from independent fragments — different functional modules designed separately, then combined into one chromosome. This is the practical assembly method for the blockchain genome pipeline.

4. Genomic Ledgers — Blockchain-Like Structures in DNA (2025)

GenomicLedgers-ExploringthePossibilityofBlockchain-LikeStructuresinDNA.pdf

Maps blockchain principles directly onto genome architecture: tandem repeats as block delimiters, DNA secondary structures and methylation as integrity-checking hashes,

retrotransposons as chain-linking references, telomere shortening and CRISPR spacer arrays as biological timestamps. Argues non-coding “junk” DNA may encode ledger-like self-validation. **Focus on:** This is the theoretical foundation — DNA already has blockchain architecture. The synthesis protocol would be engineering what nature already does.

5. JCVI-syn3.0 — Stripped Bare (2016)

JCVI-syn3.0 A synthetic genome stripped bare.pdf

Review of the JCVI-syn3.0 minimal cell construction — 531 kb, 473 genes, the smallest self-replicating organism. Discusses the modular 8-segment design where segments could be mixed and matched. **Focus on:** The modular segment approach as a precursor to block-based assembly, and why their top-down method required 4 design-build-test cycles.

6. JCVI-syn3.0 — Original Science Paper (2016)

JCVI_syn3_minimal_cell_2016.pdf

The primary research paper. Key finding: 149 of 473 genes (31%) have **unknown function** — even in the most minimal cell possible, the central planner does not know what a third of the parts do. Naive gene-by-gene knockouts failed because of “synthetic lethal pairs” (individually dispensable, collectively essential). **Focus on:** The 31% unknown genes — this is the Layer 2b argument in practice. The planner hits a knowledge ceiling. Also the synthetic lethality problem — you cannot evaluate blocks in isolation.

7. Blockchain Biology (2020)

blockchain_biology_frontiers_2020.pdf

Proposes using blockchain architecture to model cancer clonal evolution (each block = a time-stamped single-cell omics snapshot) and smart contracts to encode Boolean logic gates in synthetic biology circuits. **Focus on:** The conceptual bridge — blockchain is not just for data storage, it can model biological processes directly. The cancer evolution tracking idea parallels how a genome ledger would track assembly history.

8. Engineering Microbial Consortia by Division of Labor (2019)

engineering_microbial_consortia_division_labor_2019.pdf

Reviews how splitting complex pathways across multiple strains overcomes the “metabolic cliff” — the performance crash when one cell carries too much. Covers successful industrial examples (lignocellulose-to-biofuel, flavonoids, taxanes) and key challenges: population instability, cheater proliferation, intermediate dilution, transport barriers. **Focus on:** The practical design principles — inoculation ratios, cross-feeding engineering, biosensors for population control. These are the engineering constraints the blockchain consensus mechanism needs to handle.

9. Even Allocation Stabilizes Community (2022)

even_allocation_stabilizes_community_2022.pdf

Develops a theoretical framework showing that community stability requires **balanced benefit allocation** across cooperating strains. If one strain captures too much benefit, the consortium collapses. Validated experimentally with naphthalene-degrading consortia. **Focus on:** This is the economic principle — “voluntary trade” requires equitable exchange. Directly connects to the Layer 1b finding (Gini = 0.0, no gatekeeper).

10. MAGE — Combinatorial Genome Engineering (Sandoval et al., 2012)

sandoval-et-al-2012-strategy-for-directing-combinatorial-genome-engineering-in-escherichia-coli.pdf

Two-step strategy: TRMR maps gene-to-trait relationships genome-wide, then MAGE builds combinatorial RBS mutation libraries targeting the top hits. Applied to acetate tolerance, hydrolysate tolerance, and low pH — 25+ mutations improving growth 10-200%. Key finding: **epistatic interactions** mean beneficial single mutations do not always combine predictably. **Focus on:** The epistasis problem — blocks that work individually may conflict when combined. The compatibility engine needs to account for this, not just pairwise compatibility.

11. DOLMN — Designing Metabolic Division of Labor (Thommes et al., 2019)

thommes-et-al-2019-designing-metabolic-division-of-labor-in-microbial-communities.pdf

Mixed-integer linear programming framework that systematically explores how metabolic labor can be split across consortium members. Found counterintuitive splits (e.g., dividing the TCA cycle in half) and identified regions where obligate cross-feeding pairs **outcompete** independent single strains. **Focus on:** The computational method for deciding which blocks belong in which node — this is the FBA shadow price engine for deciding assembly order and distribution.

Recommended Reading Order

Order	Paper	Why
1	Genomic Ledgers	The theory — DNA already has blockchain architecture
2	JCVI-syn3.0 Science paper	The baseline to improve on — top-down central planning
3	Cell-free genome assembly	The assembly method — modular one-pot from fragments
4	L-SCRaMbLE	The rearrangement/mining mechanism
5	Iterative SCRaMbLE	Repeated mining with selection
6	DOLMN (Thommes)	The computational framework for distributed splits
7	Microbial consortia (Roell)	Practical consortium design principles
8	Even allocation	The economic stability requirement
9	MAGE (Sandoval)	The epistasis warning
10	Blockchain Biology	The computational bridge